

ATOPIC DERMATITIS

AT A GLANCE

- Prevalence peak of 15% to 20% in early childhood in industrialized countries
- Chronic or chronically relapsing inflammatory skin disorder characterized by:

- Pruritus

- **Eczematous dermatitis** (acute, subacute, or chronic) with typical morphology and age-specific patterns

- Facial and extensor involvement in infancy

- Flexural eczema and lichenification in children and adults

- Commonly associated with:
 - Personal or family history of atopy (asthma, allergic rhinitis, atopic dermatitis)
 - **Xerosis** and skin barrier dysfunction
 - **Immunoglobulin E (IgE) reactivity**
- Multifactorial etiology involving genetic predisposition, environmental influences, immune dysregulation, and skin barrier defects
- Associated alterations in T-cell responses, antigen processing, inflammatory cytokines, host defense proteins, allergen sensitivity, and susceptibility to infection

ETIOLOGY AND PATHOGENESIS

Atopic dermatitis (AD) is a highly pruritic inflammatory skin disease resulting from complex interactions between:

- **Genetic susceptibility** leading to defective skin barrier function

- **Dysregulation of the innate and adaptive immune systems**
- **Environmental triggers**

Risk Factors and the Hygiene Hypothesis

The rise in atopic diseases has been linked to aspects of the so-called "Western lifestyle," including:

- **Small family size**
- **Higher income and education levels (in both white and Black populations)**
- **Migration from rural to urban environments**
- **Increased use of antibiotics**

These factors have contributed to the **hygiene hypothesis**, which proposes that reduced exposure to infectious agents in early childhood—particularly through unhygienic contact with older siblings—may impair immune system development and increase susceptibility to allergic diseases.

Decreased Skin Barrier Function

AD is associated with significant impairment of the epidermal barrier due to:

- **Downregulation of cornified envelope genes** (e.g., *filaggrin*, *loricrin*)
- **Reduced ceramide levels** in the stratum corneum
- **Increased activity of endogenous proteolytic enzymes**
- **Enhanced trans-epidermal water loss (TEWL)**

External factors exacerbate barrier dysfunction:

- **Soaps and detergents** increase skin pH, activating endogenous proteases and further degrading the barrier
- **Exogenous proteases** from *Staphylococcus aureus* and house dust mites can damage the epidermis
- Atopic skin lacks sufficient **endogenous protease inhibitors**, worsening the imbalance

These abnormalities promote:

- Increased penetration of allergens through the skin
- Enhanced microbial colonization (especially by *S. aureus*)
- **Epicutaneous sensitization** to allergens, which may lead to stronger allergic immune responses compared to systemic or airway exposure
- Potential predisposition to respiratory allergies later in life

Immunopathology of Atopic Dermatitis

Skin Immune Environment

- **Clinically unaffected skin** in AD patients shows: - Mild epidermal hyperplasia
- Sparse perivascular T-cell infiltrate
- **Acute lesions** are marked by: - **Spongiosis** (intercellular edema in the epidermis)

Antigen-Presenting Cells and IgE

- Dendritic antigen-presenting cells—including **Langerhans cells (LCs)** and macrophages—in lesional and nonlesional skin display **surface-bound IgE**
- This IgE binding enhances allergen capture and presentation, contributing to Th2-driven inflammation
- Results in heightened immunologic responses to allergens and microbial antigens

TABLE 14-1: Features of Atopic Dermatitis

Major Features

- Pruritus

- Rash on face and/or extensors in infants and young children
- Lichenification in flexural areas in older children and adults
- Chronic or chronically relapsing course
- Personal or family history of atopic diseases (asthma, allergic rhinitis, atopic dermatitis)

Other Common Findings

- Xerosis (dry skin)
- Dennie-Morgan folds (prominent infraorbital skin folds)
- Allergic shiners (periorbital darkening)
- Facial pallor
- Pityriasis alba
- Keratosis pilaris
- Ichthyosis vulgaris
- Hyperlinearity of palms and soles
- White dermatographism (white line after skin stroking)
- Conjunctivitis
- Keratoconus
- Anterior subcapsular cataracts
- Elevated serum IgE
- Immediate-type skin test reactivity to allergens